

CO2 AT 1 Scenario Based questions
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QUESTIONS

- Q1**
Student Marks Analysis
10 marks
- Q2**
Attendance Tracker
10 marks
- Q3**
Exam Result System
10 marks
- Q4**
Duplicate Names Detection
10 marks
- Q5**
Bank Transaction Summary
10 marks

5/5 answered

Q1 Student Marks Analysis

10 marks

A teacher has stored the marks of 10 students in an array. She wants to analyze the performance of the class by finding the highest mark, calculating the average mark, and counting how many students scored above the average.

Sample Input:

Marks = {45,67,89,32,76,50,91,84,60,73}

Sample Output:

Highest = 91

Average = 66.7

Students Above Average = 5

Test Cases:

- {10,20,30,40,50,60,70,80,90,100}
- {50,50,50,50,50,50,50,50,50,50}
- {100,99,98,97,96,95,94,93,92,91}
- {0,0,0,0,0,0,0,0,0,0}
- {35,45,55,65,75,85,95,100,20,10}

Your Code

```

1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = 10;
6         int[] marks = new int[n];
7         for (int i = 0; i < n; i++) {
8             marks[i] = sc.nextInt();
9         }
10        int highest = marks[0];
11        int sum = 0;
12        for (int i = 0; i < n; i++) {
13            if (marks[i] > highest) {
14                highest = marks[i];
15            }
16            sum += marks[i];
    
```

Input

45 67 89 32 76 50 91 84 60 73

Output

```

Highest = 91
Average = 66.7
Students Above Average = 5
    
```

Visible Test Cases

Test Case 1

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Q2 Attendance Tracker

10 marks

A school records attendance using a string such as 'PPAAPPPAPP'. Count Present, Absent, attendance percentage and display 'Not Eligible' if below 75%.

Sample Input:

PPAAPPPAPP

Sample Output:

Present=7 Absent=3 Attendance=70% Not Eligible

Test Cases:

1. PPPPPPPPP
2. AAAAAAAAAA
3. PPAPPPAPPA
4. PAPAPAPAPA
5. PPPAAPPPAA

Your Code

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String s = sc.nextLine();
6         int present = 0, absent = 0;
7         for (char c : s.toCharArray()) {
8             if (c == 'P') present++;
9             else if (c == 'A') absent++;
10        }
11        int total = present + absent;
12        double percentage = (present * 100.0) / total;
13        StringBuilder sb = new StringBuilder();
14        sb.append("Present=").append(present)
15          .append(" Absent=").append(absent)
16          .append(" Attendance=").append((int) percentage).append("%");
17        if (percentage < 75) {
18            sb.append(" Not Eligible");
19        }
20    }
21 }
```

Input

PPAAPPPAPP

Output

Present=7 Absent=3 Attendance=70%
Not Eligible

Visible Test Cases

Test Case 1

Test Case 2

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5/5 answered

Q3 Exam Result System

10 marks

A teacher stores student marks in an array. Count pass/fail and assign grades (A,B,C,Fail).

Sample Input:
{45,67,89,32,76,50}

Sample Output:
Pass=4 Fail=2

Test Cases:
1. {90,91,92}
2. {75,80,89}
3. {50,60,74}
4. {49,48,20}
5. {100,75,50,49}

Your Code

```
1 import java.util.*;  
2 public class Main {  
3     public static void main(String[] args) {  
4         Scanner sc = new Scanner(System.in);  
5         List<Integer> a = new ArrayList<>();  
6         while (sc.hasNextInt()) a.add(sc.nextInt());  
7         int pass = 0, fail = 0;  
8         for (int x : a) {  
9             if (x >= 50) pass++;  
10            else fail++;  
11        }  
12        System.out.println("Pass=" + pass + " Fail=" + fail);  
13    }  
14 }
```

Input

100 75 50 49

Output

Pass=3 Fail=1

Visible Test Cases

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Test Case 3

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5/5 answered

Q4 Duplicate Names Detection

10 marks

An organization maintains employee names in an array. Identify duplicate names and count occurrences.

sample input:

{Ram,Sita,Ram,John,Sita}

Sample Output:

Ram=2 Sita=2

Test Cases:

1. {A,B,C}
2. {Tom,Tom,Tom}
3. {X,Y,X,Z,Y}
4. {A,A,B,B,C,C}
5. {Raj,Raj,Raj,Raj}

Your Code

```
1 import java.util.*;  
2  
3 public class Main {  
4     public static void main(String[] args) {  
5         Scanner sc = new Scanner(System.in);  
6         sc.useDelimiter("\\s+");  
7  
8         Map<String, Integer> map = new LinkedHashMap<>();  
9         while (sc.hasNext()) {  
10             String s = sc.next();  
11             if (!s.isEmpty()) {  
12                 map.put(s, map.getOrDefault(s, 0) + 1);  
13             }  
14         }  
15         sc.close();  
16  
17         StringBuilder sb = new StringBuilder();  
18         for (Map.Entry<String, Integer> e : map.entrySet()) {  
19             if (sb.length() > 0) sb.append(' ');  
20             sb.append(e.getKey()).append('=').append(e.getValue());  
21         }  
22         System.out.println(sb);  
23     }  
24 }
```

Input

RAJ RAJ RAJ RAJ

Output

RAJ=4

Visible Test Cases

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Test Case 3

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3/5 answered

Q5 Bank Transaction Summary

10 marks

A bank stores transactions where positive values are deposits and negative values are withdrawals. Calculate final balance and count deposits and withdrawals.

Sample Input:
{500,-200,1000,-300,700}

Sample Output:
Balance=1700 Deposits=3 Withdrawals=2

Test Cases:
1. {100,-50}
2. {-100,-200}
3. {100,200,300}
4. {0,0,0}
5. {100,-100,50,-25}

Your Code

```
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         List<Integer> nums = new ArrayList<>();
8         while (sc.hasNextInt()) nums.add(sc.nextInt());
9
10        long balance = 0;
11        int deposits = 0, withdrawals = 0;
12
13        for (int x : nums) {
14            balance += x;
15            if (x > 0) deposits++;
16            else if (x < 0) withdrawals++;
17        }
18
19        System.out.print("Balance=" + balance + " Deposits=" + deposits + " Withdrawals=" + withdrawals);
20    }
21 }
```

Input

100 -100 50 -25

Output

Balance=25 Deposits=2 Withdrawals=1

Visible Test Cases

Test Case 1

Test Case 2